

ROBERT A. COCHRAN III

rac@cs.unc.edu — 864.680.9876 — robbycochran.com

OBJECTIVES

Seeking full-time work that allows me to continue researching, designing and implementing novel solutions for safe, secure and robust networked and distributed systems; particularly interested in applied formal methods, automated program analysis and software correctness.

EDUCATION

UNIVERSITY OF NORTH CAROLINA AT CHAPEL HILL Chapel Hill, NC

Doctor of Philosophy in Computer Science

2016

Thesis: *Symbolic Verification of Remote Client Behavior in Distributed Systems*

Advisor: Prof. Michael K. Reiter

CLEMSON UNIVERSITY Clemson, SC

Bachelor of Science in Computer Science *Magna Cum Laude*

2006

EMPLOYMENT AND RESEARCH EXPERIENCE

UNIVERSITY OF NORTH CAROLINA AT CHAPEL HILL Chapel Hill, NC

Research Assistant with Prof. Michael Reiter.

2008 - 2016

- *Cheat Detection in Online Games*: Led development of an award-winning security method called symbolic client verification; demonstrated and implemented a framework to detect player misbehavior in the network logs of online games [1, 5]
- *Data-Driven Symbolic Execution*: Investigated the use of instruction traces to guide symbolic execution in the context of client behavior verification; designed and implemented a toolchain using clustering and novel edit distance metrics that improved performance significantly [4]
- *Parallel Symbolic Execution*: Refactored core components of the KLEE sequential symbolic execution engine to enable thread-level parallelism; developed a lock-free algorithm leading to high performance on many-core architectures [2]
- *Verification of Cryptographic Clients*: Designed a multi-pass approach to enable verification of network behavior in cryptographic clients using symbolic execution; implemented extensions to enable detection of OpenSSL client misbehavior (e.g. HeartBleed attacks) [2]

Research Assistant with Prof. Ketan Mayer-Patel.

2007- 2008

- *Spline-based Image Compression*: Investigated an image compression technique based on image decomposition using L1 splines; implemented the technique to evaluate image quality, compression ratio and speed

Teaching Assistant

Fall 2008

- *COMP 535 Introduction to Computer Security*: Graded student programs and written assignments; led recitations and exam review sessions

MICROSOFT RESEARCH Redmond, WA

Research in Software Engineering (RiSE) Intern with Dr. Ben Livshits

Summer 2013

- *Program Synthesis via Crowd-sourcing*: Explored approaches for improving regular expressions using the "wisdom of crowds"; designed and implemented a toolchain that utilized genetic programming, symbolic automata and on-demand online workers [3, 7]

INTEL CORPORATION Hillsboro, OR

Software Engineer in the Visual Computing Group

2006-2007

- *GPU Hardware Development*: Designed and developed a tool to interpret and record for playback Direct3D and OpenGL API calls; tool was crucial in performance analysis of prototype graphics hardware

CLEMSON UNIVERSITY Clemson, SC

Independent Research with Prof. Robert Geist

2006

- *Rendering with Realistic Lighting*: Designed and implemented a GPU-based algorithm for computing an approximation of indirect illumination [6]

Undergraduate Research Assistant with Prof. Jim Martin

2005

- *Network Data Analysis*: Developed a toolchain to evaluate updates to the DOCSIS MAC layer in network simulation research

PUBLICATIONS

JOURNAL ARTICLES

- [1] D. Bethea, **R. Cochran** and M. K. Reiter. "Server-side verification of client behavior in online games," In *ACM Transactions on Information and System Security (TISSEC)* 14(4), December 2011.

PEER-REVIEWED CONFERENCE PROCEEDINGS

- [2] A. Chi, **R. Cochran**, M. Nesfield, M. K. Reiter and C. Sturton. "A System to Verify Network Behavior of Known Cryptographic Clients," In *Proceedings of the 14th USENIX Symposium on Networked Systems Design and Implementation (NSDI)*, March 2017. To appear.
- [3] **R. Cochran**, L. D'Antoni, B. Livshits and M. Veanes. "Program Boosting: Program Synthesis via Crowdsourcing," In *Proceedings of the 42nd ACM SIGPLAN-SIGACT Symposium on Principles of Programming Languages (POPL)*, January 2015.
- [4] **R. Cochran** and M. K. Reiter. "Toward online verification of client behavior in distributed applications," In *Proceedings of the 20th ISOC Network and Distributed System Security Symposium (NDSS)*, February 2013.
- [5] D. Bethea, **R. Cochran** and M. K. Reiter. "Server-side verification of client behavior in online games," In *Proceedings of the 17th ISOC Network and Distributed System Security Symposium (NDSS)*, February 2010.
- [6] **R. Cochran** and J. Steele. "Second-order illumination in real-time," In *Proceedings of the ACM Southeast Regional Conference (ACMSE)*, March 2007.

PATENTS

- [7] B. Livshits and **R. Cochran**. "Program boosting including using crowdsourcing for correctness" US Patent App. 14/212,462, Sept 17, 2015.

AWARDS AND DISTINCTIONS

Best Paper Award, Network and Distributed System Security Symposium [2]	2010
Best Paper Award, Annual ACM Southeast Regional Conference [3]	2007
Outstanding Senior in Computer Science, Clemson University	2006
Donald A. Norton Computer Science Scholarship, Clemson University	2006
Upsilon Pi Epsilon Honor Society, Clemson University	2005
United States Senate Page	2000
Eagle Scout Award	1999

SERVICE

EXTERNAL REVIEWER

ACM Conference on Data and Application Security and Privacy	2013, 2014
Network & Distributed System Security Symposium	2013, 2014

TALKS AND PRESENTATIONS

Conference Talk: <i>Toward online verification of client behavior in distributed applications.</i> [4]	
20th ISOC Network and Distributed System Security Symposium, San Diego, CA.	February 2013
Conference Talk: <i>Server-side verification of client behavior in online games.</i> [5]	
17th ISOC Network and Distributed System Security Symposium, San Diego, CA.	February 2010
Tutorial: <i>Introduction to General Purpose GPU (GPGPU) Programming</i>	
45th Annual ACM Southeast Regional Conference, Winston-Salem, NC.	March 2007

EXPERTISE

PROGRAMMING:	C, C++, C++11, C#, Java, Python, R, Bash
TOOLS AND PLATFORMS:	LLVM, Git, L ^A T _E X, VTune, SMT/SAT solvers, Mechanical Turk, Raspberry Pi
COMMUNICATION:	Outstanding presentation skills and excellent writing ability
HOBBIES:	Improv comedy performance, analog synthesizer design, mountain biking